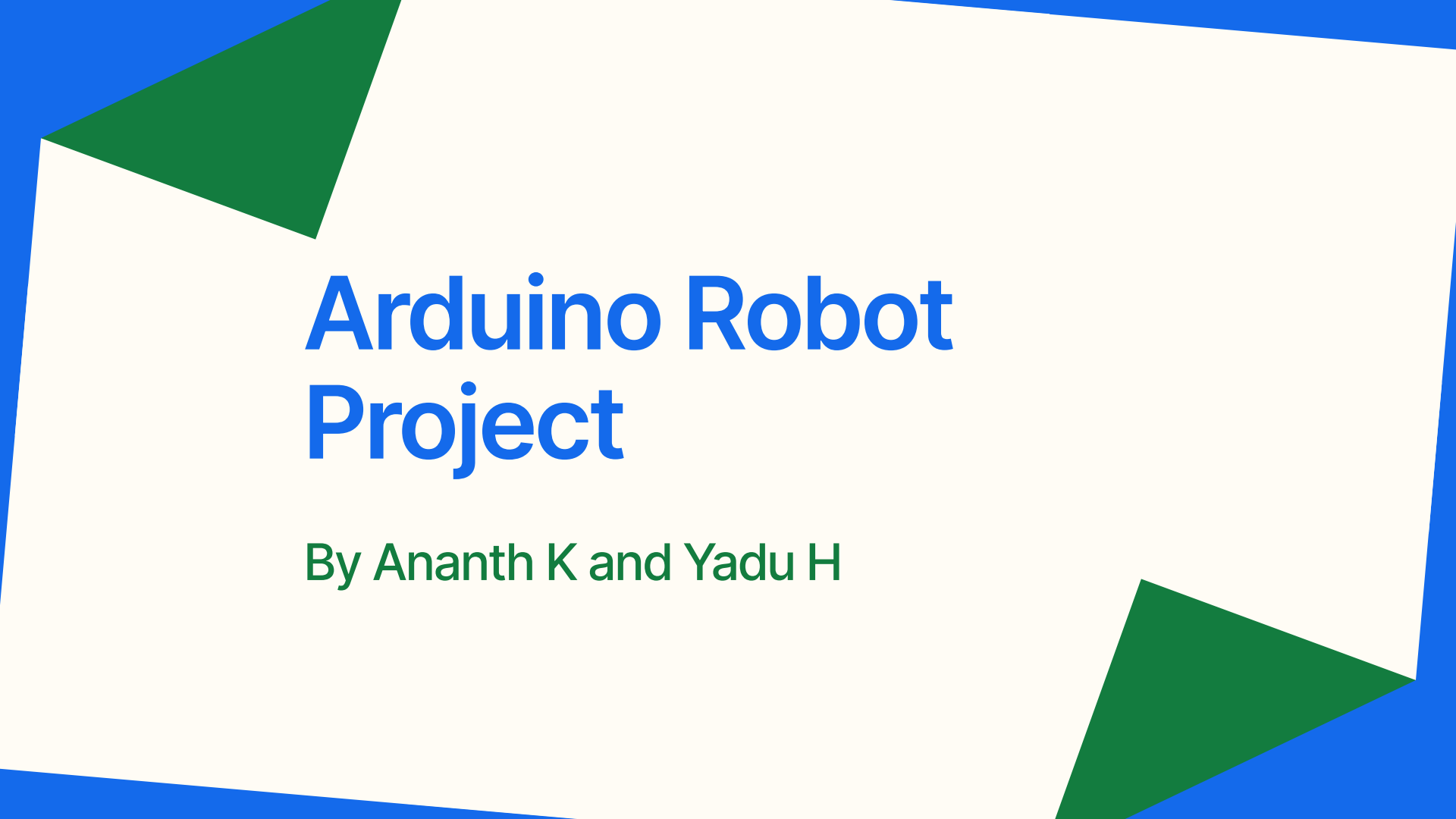
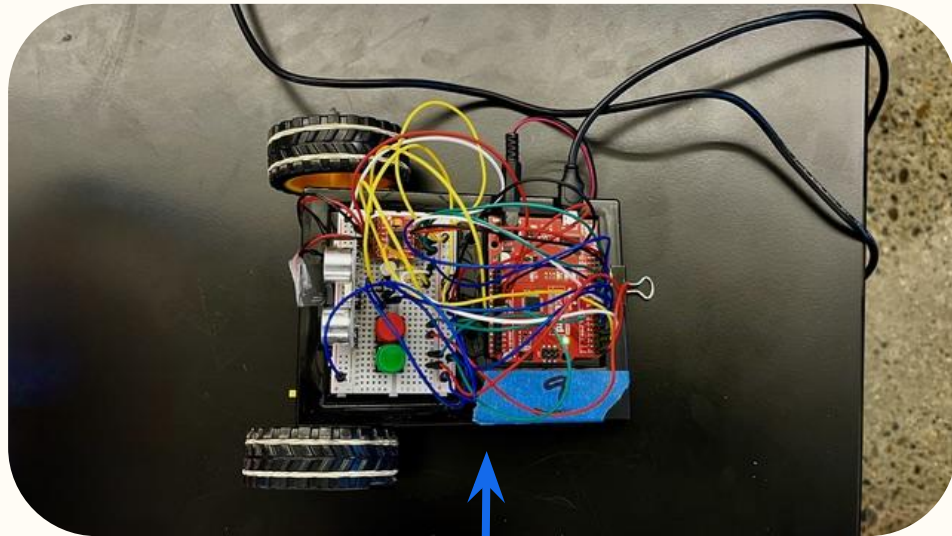




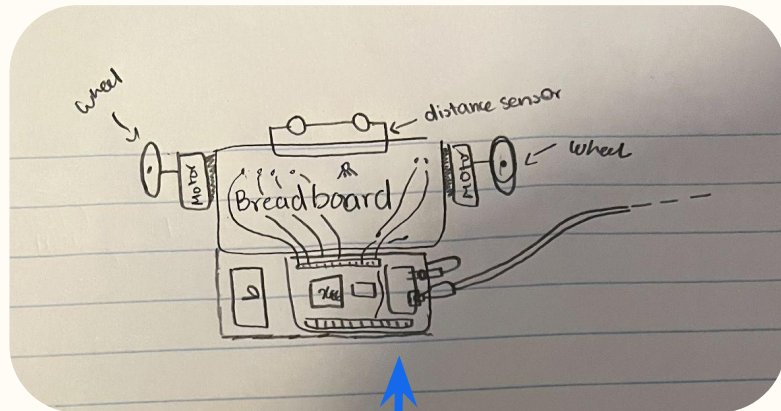
# Arduino Robot Project

By Ananth K and Yadu H

# Meet Bob: Our Robot

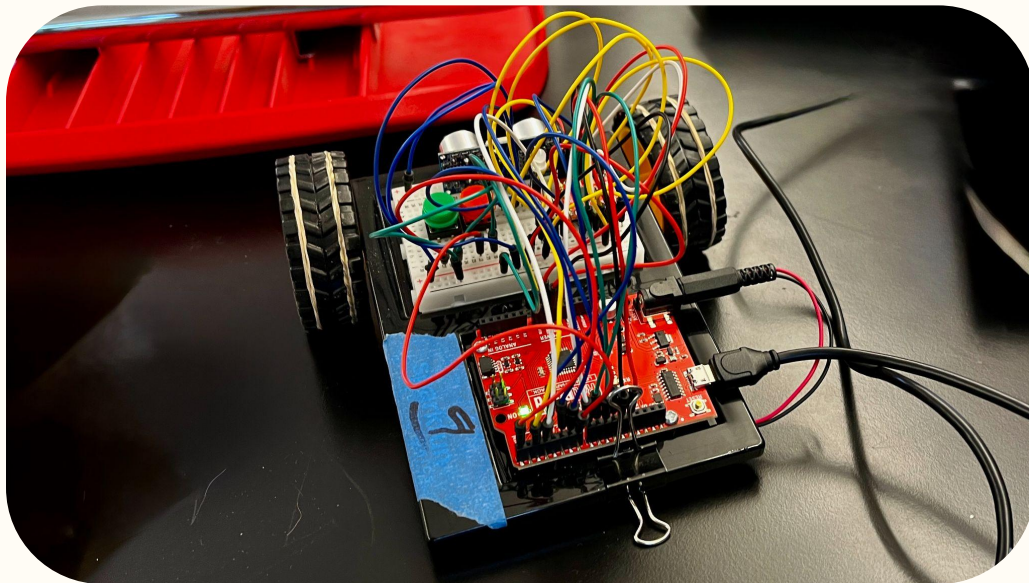


Bob



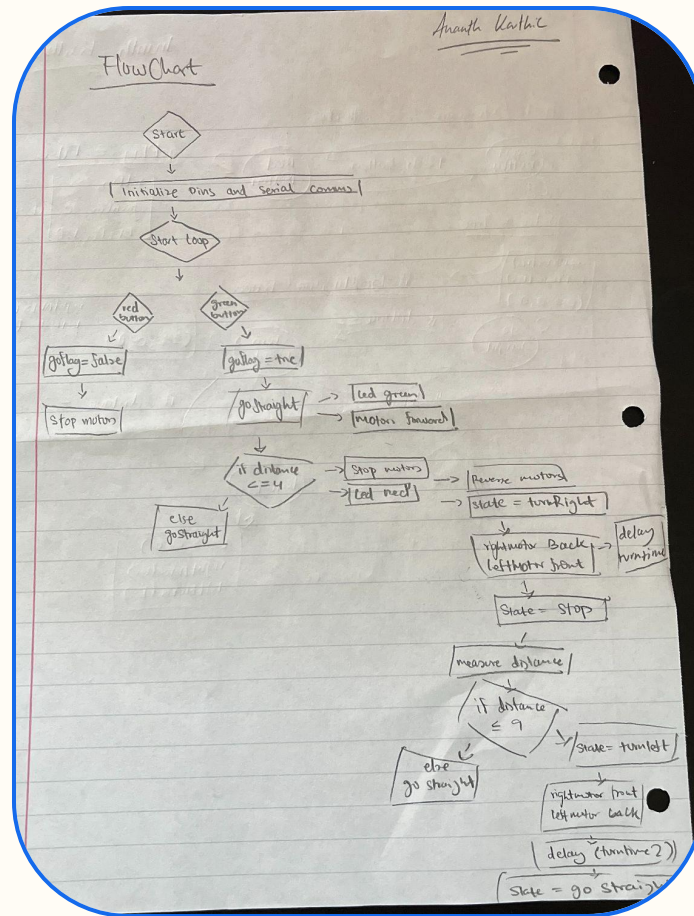
Diagram

# System Basics



- **Purpose:** Make the robot navigate autonomously through the maze
- **Performance Requirements:**
  - 3 tries to complete the maze
  - 90 seconds
- **Components:**
  - 32 Jumper Wires, motor driver, 2 gear motors, 2 wheels, RGB LED, Rubber bands, arduino board, red board, tape, red button, green button, Distance sensor

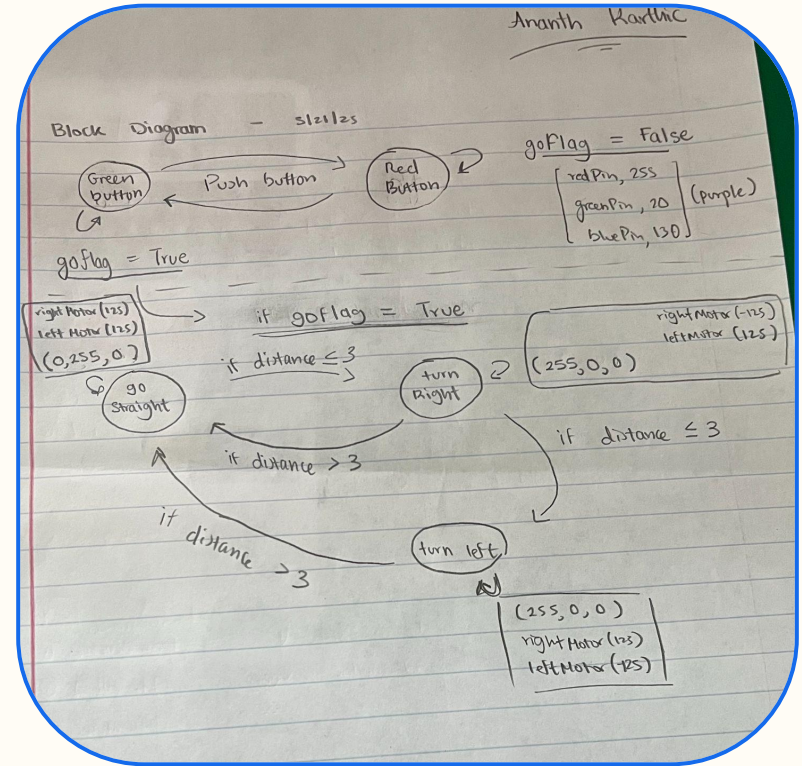
- When the **red button** is pressed:



# State Machine

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- Our robot is controlled by a **State Machine**
- **goStraight:**
  - The robot moves forward. If it detects an obstacle closer than 4 cm, it stops, backs up, and switches to the turnRight state.
- **turnRight:**
  - The robot turns right for a set duration, then moves to the stop state.
- **stop:**
  - The robot slowly moves forward, checks for obstacles again. If still blocked (distance  $\leq 9$  cm), it backs up and turns left. Otherwise, it resumes going straight.
- **turnLeft:**
  - The robot turns left for a set duration and then resumes moving forward by switching back to goStraight.



Loop:

If green button is pressed:

Enable movement (`goFlag = true`)

Else if red button is pressed:

Disable movement (`goFlag = false`)

Set LED to purple

If movement is enabled:

If state is "goStraight":

Move forward

Set LED to green

If obstacle is very close:

Stop

Set LED to red

Back up for a while

Set state to "turnRight"

Else if state is "turnRight":

Turn right

Wait

Set state to "stop"

Else if state is "stop":

Move forward slowly

Wait and recheck distance

If obstacle still close:

Back up

Set state to "turnLeft"

Else:

Set state to "goStraight"

Else if state is "turnLeft":

Turn left

Wait

Set state to "goStraight"

Else:

Stop all movement

Wait before next loop



# Summary

Overall, building the robot was pretty straightforward, but we ran into some issues getting it to work consistently. We had a few wiring problems that slowed us down, but we managed to fix them and get back on track. The coding part was simple since it was similar to Python, which made it easier to program the robot. As the demo gets closer, Yadu and I do wish we had more time to test in the maze. It's been hard to get access with so many people making last-minute changes. If we had more testing time, the number of people waiting would probably go down. In the end, this was a really fun unit. I would definitely recommend it to future students because it teaches a lot about electronics and makes the learning experience more interactive.